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Aldrin Joan Pandian W

Programming Languages: Java, C++, Python

Mobile Development: Android Development, Flutter

Machine Learning & AI: Natural Language Processing (TF-IDF, LDA, NMF, Semantic Search), Machine Learning, Generative AI (GPT-5, LLM fine-tuning), Multi-Agent Systems, Prompt Engineering, Model Context Protocol Server.

Web Development: Web Development (React JS, Next JS), API Integration, JavaScript, WebSocket.

Cloud & DevOps: Azure Deployment, Cloud-based AI applications

Tools & Frameworks: LangChain, LangGraph, LangSmith, YOLO model, Responsive Design, Real-time Data Handling, File Processing, Automation, FastMCP, IDE Extensions.

Certification:

- EC Council Certified Ethical Hacking V12 – EC Council (2023-27)
- AI Engineering – IBM

EDUCATION

Board	Tenure	Educational Institution	CGPA/Percentage
B. Tech (CSE) AI&ML	September 4, 2023 - Ongoing	VIT Bhopal University, Bhopal	8.63/10
Class XII	May 10, 2023	Carmel Garden Matric Hr Sec School, Coimbatore	75.0%
Class X	May 28, 2021	Carmel Garden Matric Hr Sec School, Coimbatore	100%

ACADEMIC PROJECTS

End-to-end
Developer

▪ GraviGauge – VS Code Extension (Antigravity Quota Monitor) (2026)

- **Description:** Designed and developed a production-ready VS Code extension that monitors real-time AI model quota usage from the local Antigravity application. Reverse-engineered Antigravity's internal process, authentication flow, and API behavior to enable zero-configuration, secure local integration. **(776 Downloads in Antigravity)**

- Features:

- Real-Time Monitoring — Displays lowest available model quota directly in VS Code status bar
- Dynamic Port Discovery — Automatically detects the running Antigravity process and resolves server port
- Session Token Authentication — Extracts and reuses active session tokens (no manual API key required)
- Auto & Manual Refresh — Background polling (60s) + on-click refresh
- Quota Intelligence — Categorizes model health (healthy, low, exhausted) with visual indicators
- Secure Local Networking — Handles localhost SSL bypass for self-signed certificates
- Detailed Tooltip UI — Shows full model breakdown (Gemini, Claude, GPT variants, etc.)
- Robust Error Handling — Handles process detection failures, zero quotas, and connection states

- Engineering Highlights:

- Reverse engineered Antigravity local server communication flow
- Implemented process scanning via PowerShell to dynamically locate active instances
- Built lightweight status bar UI using VS Code Extension API
- Designed efficient polling mechanism with minimal performance overhead
- Structured modular TypeScript architecture for maintainability

- Technology:

- TypeScript, Node.js, VS Code Extension API, PowerShell Process Detection, Local HTTPS Handling, REST API Integration

- Role:

- Sole Developer — Architecture, Reverse Engineering, API Integration, UI/UX, Packaging, Publishing

- Publishing:

- Published on Microsoft VS Code Marketplace
 - <https://marketplace.visualstudio.com/items?itemName=AldrinJoanPandianW.gravigauge>

	- Published on Open VSX Registry https://open-vsx.org/extension/AldrinJoanPandianW/gravigauge - Complete extension packaging, semantic versioning, and release management
End-to-end Developer	▪ Bollywood Bias Buster(June 30, 2025 – June 19, 2025) - Description: Detect, quantify, and remediate gender bias in Bollywood media. Classifies script dialogues by stereotype, rewrites biased text, visualizes bias trends, and generates structured PDF/CSV reports. Extended to posters (LLaVA-1.5) and trailers (emotion–gender plots) with Wikipedia-based linguistic and network analytics. - Features: ▪ Data Cleaning — Extract gender-tagged dialogues from scripts (PDF → CSV) using NLP-based character and gender identification ▪ Classification — Detect gender stereotypes with local Mistral-7B (GGUF) ▪ Visualization — Bias category distribution, gender-wise trends ▪ Rewrite — Stereotype-free rewriting using LLM guidance ▪ Report Generation — Downloadable PDF + CSV reports ▪ Poster Analysis — Visual bias detection with LLaVA-1.5 ▪ Trailer Analysis — Emotion–gender timeline plots ▪ Wikipedia Analytics — Adjective/verb usage, centrality, coreference, songs metadata - Technology: Streamlit, Hugging Face Transformers (Mistral-7B, LLaVA-1.5), PyMuPDF, Pandas, Matplotlib, Seaborn, NetworkX, NLP pipelines - Team Project: Individual - Role: End-to-end Developer — bias classification, multimodal integration, UI/UX, and report generation - GitHub: https://github.com/Aldrin-Joan/Bolly_Bias_Streamlit_By_Aldrin_Joan_Pandian.git
Android App & Large File Processing using AI & ML	▪ Legal AI Android & Web App (March 29, 2025 – Ongoing) - Description: Developed a cross-platform AI legal assistant for Android and web. Enabled users to post legal issues, get AI-generated insights, access legal documents, and receive real-time legal updates with misinformation checks. Included voice-based multilingual chatbot and clause analysis. - Technology: React.js, Flutter, Django, FastAPI, MongoDB, Pegasus Indian Legal, CUAD, SpaCy, NLTK, Google STT/TTS, TD-IDF + LDA, AWS, Firebase, LangGraph - Team Project: 4 Members - Role: Front End (ReactJS), Flutter App Developer and Large File Processing using AI & ML
Flutter Developer	▪ RehabEase – AI-Powered Rehabilitation Platform (Dec 20, 2024 – May 26, 2025) - Description: Designed a mobile and web app for guided rehab exercises. Integrated motion tracking, real-time feedback, and progress monitoring. - Technology: Python Flask, Dart, Flutter, Firebase and Computer Vision - Team Project: 3 Members - Role: Team Leader and Android and IOS App Developer
Front End	▪ Deep Learning-Based Knee Osteoarthritis Diagnosis (Jan 29, 2025 – April 14, 2025) - Description: Implemented CNN and ensemble models to classify OA severity from X-rays. Added Grad-CAM and a web interface for real-time diagnosis. - Technology: Python, CNN, ReactJS, Machine Learning - Team Project: 6 Members - Role: Front End Developer
Model Training	▪ Heart Disease Prediction Using Machine Learning (Sep 23, 2024 – Dec 18, 2024) - Description: Created a machine learning model to predict heart disease risk. Applied feature engineering, classification, and visualization techniques. - Technology: Python, Jupyter, Machine Learning - Team Project: 2 Members - Role: Team Leader and Model Training
Model Training	▪ Hand Sign Language Translation Using CNN (Oct 21, 2024 – Dec 18, 2024) - Description: Built a CNN-based model to convert hand signs to text and vice versa. Enhanced gesture recognition using computer vision for accessibility. - Technology: Python and CNN Model - Team Project: 5 members - Role: Model Training
Front End	▪ AI-Powered Mentor-Mentee Platform (March 14, 2024 – Oct 23, 2024) - Description: Developed a web app with AI-based matching to connect mentors and mentees. Integrated real-time chat, video calls, and calendar booking. - Technology: NextJS, Django, Firebase, MySQL, Python - Team Project: 6 Members

- **Role:** Team Leader and Front-End Developer

INTERNSHIP

Docu3C (Seattle, Washington, USA) (July 21, 2025 – Till Date)	AI Engineer • Prompt Engineering & Optimization: Designed and optimized effective prompts for Large Language Models (LLMs) including GPT-5, Claude, and DeepSeek to drastically improve response accuracy, reliability, and contextual awareness. • Multi-Agent System Development: Architected and deployed intelligent, collaborative multi-agent workflows using LangChain, LangGraph, LangSmith, LiteLLM, and Temporal orchestration. • AI Research & Development (R&D): Spearheaded R&D initiatives to evaluate emerging LLM capabilities and prototype experimental multi-agent applications for specialized domain tasks. • Advanced NLP & Semantic Analysis: Applied advanced Natural Language Processing techniques (TF-IDF, LDA, NMF) for text analysis, topic modeling, and semantic/vector search capabilities. • Cloud-Based AI Deployment: Developed, optimized, and deployed scalable AI applications on Microsoft Azure, ensuring high performance, efficient resource orchestration, and seamless system integration. • Automated Workflows & Integrations: Built automated, context-aware applications for real-time data handling, complex file processing, and deep integrations with Salesforce and ServiceNow. • Agentic System Optimization: Spearheaded the debugging, optimization, and error-handling of complex agentic frameworks to ensure robust system reliability in production environments. • Protocol & API Expansion: Upgraded system function-calling capabilities by building and integrating robust Model Context Protocol (MCP) Servers. • End-to-End (E2E) Application Delivery: Acted as the sole developer to architect, build, and deploy complete AI-driven applications from the ground up, managing the full software lifecycle from initial concept to final Azure deployment. Skills • AI & Orchestration: Prompt Engineering, Multi-Agent Systems, Multi-Model Agentic Systems, AI Research & Development (R&D), LangChain, LangGraph, LangSmith, LiteLLM, Temporal. • Generative AI & LLMs: GPT-5, Claude, DeepSeek, LLM Fine-tuning, Context Management, Function Calling, Model Context Protocol (MCP) Servers. • Natural Language Processing: TF-IDF, LDA, NMF, Semantic Search, Vector Search, Text Analysis. • Cloud & Architecture: Azure Deployment, Cloud-based AI Applications, Scalable Architecture, System Orchestration. • Engineering & Automation: Salesforce & ServiceNow Automation, Real-time Data Handling, File Processing Pipeline Development, Debugging & Error Handling. • End-to-End (E2E) Application Development.
amasQIS.ai (Omen) (April 28, 2025 – October 31 st , 2025)	• Worked as part of the Front-End team for the Personal Protective Equipment Compliance Project, contributing to webpage design and UI development. • Served as a Backend Developer in the Predictive Maintenance Project, integrating the YOLO model to meet custom requirements for defect detection. • Acting as Lead Flutter Developer for two projects, • LLM Workspace Adviser and VLM Workspace Adviser • Handling end-to-end mobile application development.
ApexPlanet Software Pvt. Ltd. (Oct 20, 2024 – Dec 4, 2024)	• Completed structured tasks involving webpage creation, responsive design, JavaScript interactivity, API integration, and localStorage. • Developed a dynamic to-do list, image gallery, and an E-commerce website as the final project. • Enhanced frontend development skills and implemented real-time data handling and performance optimization techniques.

CO-CURRICULARS

Coding	▪ http://www.skillrack.com/profile/452987/63a1b8e881639ac762c3e1187e2cdf1f0f66308a ▪ 87600 Rank
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EXTRA-CURRICULARS AND ACHIEVEMENTS

Achievements	▪ Research Paper Presented, 3rd International Conference ICRP-2024 – ICRP (2024) ▪ Attended, CSIR Sponsored National Seminar on Vehicular Cloud Model – CSIR (2024)
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Responsibilities	▪ UGC NEP SAARTHI – Student Ambassador (Jan 02, 2025 – Till Date)
Extracurricular	▪ Completed 7 Grades in Keyboard – London College of Music (LCM) & Trinity College of Music

ADDITIONAL INFORMATION	
Research	▪ Research Paper Published, AI-Driven Predictive Maintenance for Smart Grid Components Architecture Applications, Challenges and Opportunities, Springer, 2024. ▪ Book Chapter Published, The Role of Artificial Intelligence in 6G Networks, Architecture, Protocol, Transmission and Applications, IGI Global, 2025. ▪ <i>Optimizing Video Compression for Real-Time and Bandwidth-Constrained Applications in IoT Enabled Cyber-Physical Systems</i> ▪ <i>Machine Learning Approaches to Support Corporate Environmental Governance through Accurate Emission Forecasting, Carbon Offset Allocation, and Green Fund Optimization</i> ▪ A Next-Generation Computing Approach for Early and Accurate Lung Cancer Detection Using an Integrated Hybrid Deep Learning Framework
Hobbies	▪ Reading Japanese Mangas ▪ Playing Games ▪ Playing Basketball
Languages	▪ English, Tamil, Hindi, Japanese